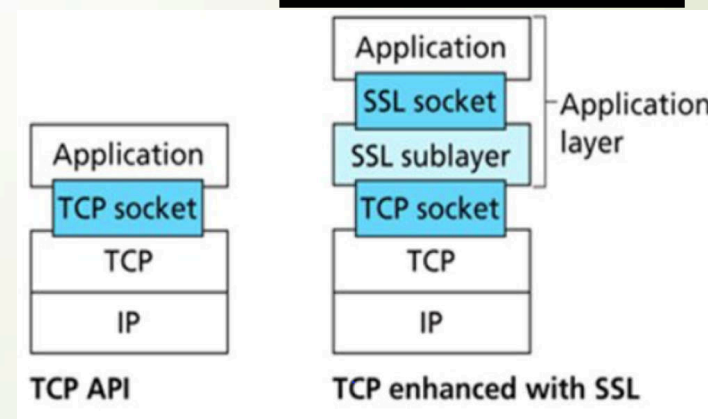
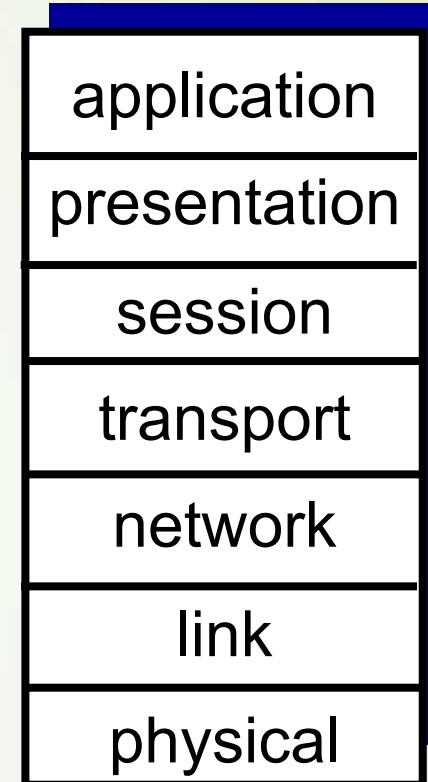


ISO/OSI 7-Layer Reference Model

- **presentation:** common data interpretation, e.g., encryption, compression, machine-specific conventions
- **session:** synchronization, checkpointing, recovery of data exchange
- Internet stack misses these layers!
 - these services, if needed, are done in **app** layer
 - Example: SSL



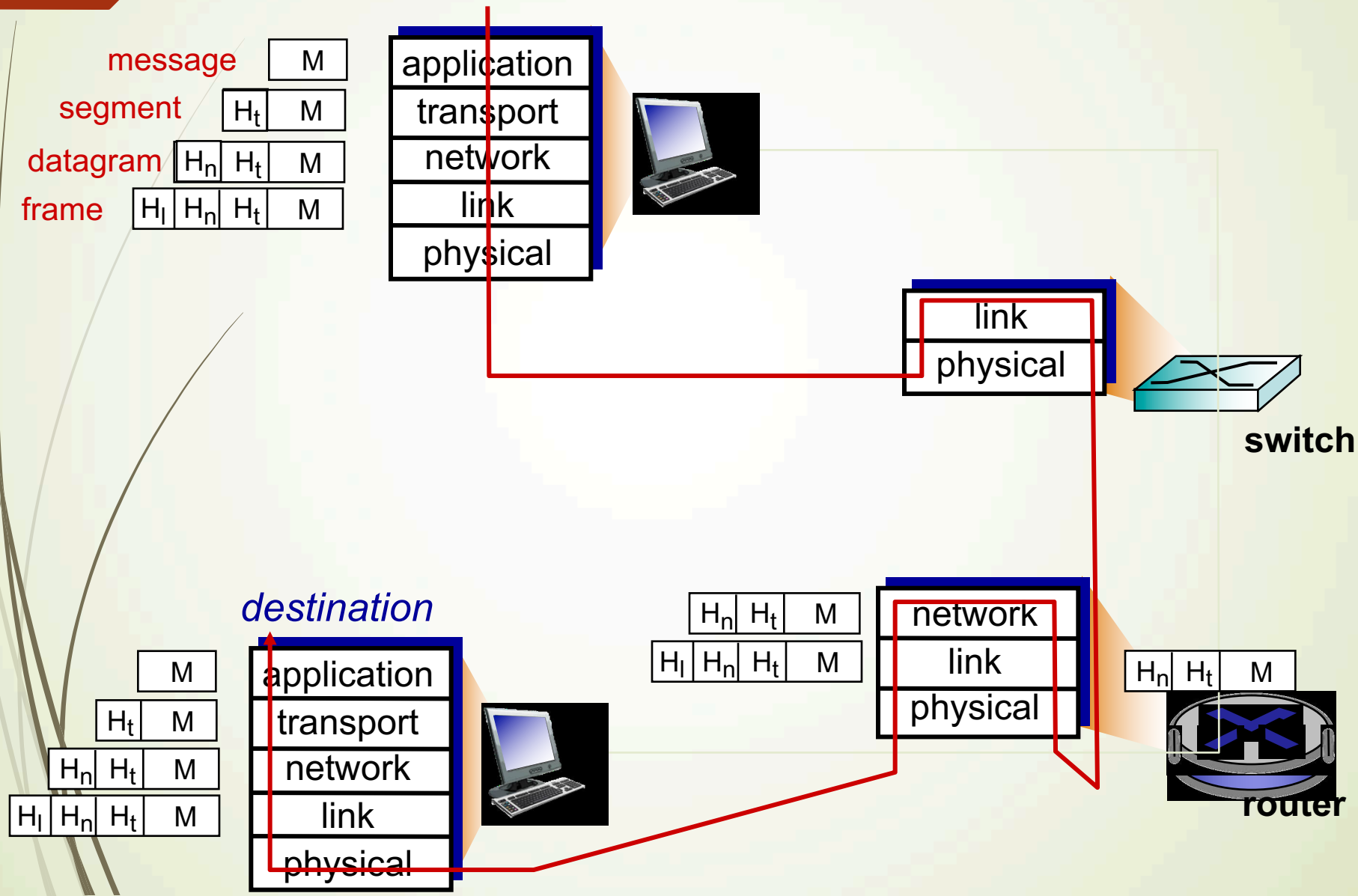
UNIX Socket API

- A socket is an abstraction of a communication **endpoint**
- Socket descriptors are implemented as **file descriptors** in the UNIX System
- General functions that deal with file descriptors, such as **read** and **write**, also work with a socket descriptor
- Creating a socket: `int socket(int domain, int type, int protocol);`
- Release a socket: `close(...)`

Sample Socket APIs

- For connection oriented (TCP) only
 - `ssize_t send(int sockfd, const void *buf, size_t nbytes, int flags);`
- For both Connection oriented (TCP) and connectionless (UDP)
 - `ssize_t sendto(int sockfd, const void *buf, size_t nbytes, int flags, const struct sockaddr *destaddr, socklen_t destlen);`
- Returns: number of bytes sent if success, or -1 on error

Encapsulation and Decapsulation of Packets



Agenda

- What is the Internet
- Network Edge
- Network Core: Circuit Switched vs. Packet Switched
- Protocol Layers and UNIX Sockets
- History and Internet design philosophy

History of the Internet

- 1961-1972: Early packet-switching principles
 - 1972: ARPAnet has 15 nodes
- 1972-1980: Internetworking, new and proprietary networks
 - 1974: Cerf and Kahn's internetworking principles defines today's Internet
 - 1979: ARPAnet has 200 nodes
- 1980-1990: new protocols, a proliferation of networks
 - 1990: 100,000 hosts
- 1990-2000's: commercialization, the Web, new apps
 - Late 1990's: 50 million hosts, 100 million+ users

2005-Present's Internet

- ▶ ~5B devices attached to Internet (2016)
 - ▶ smartphones and tablets
- ▶ More broadband and high-speed wireless access
- ▶ emergence of online social networks:
 - ▶ Facebook 2.27 billion users; Instagram 1 billion users, Twitter 336 million (late 2018).....
- ▶ Service/content providers (like Google) create their own networks
 - ▶ bypass traditional ISPs
- ▶ universities, enterprises run their services in cloud

Cerf and Kahn's Internetworking Principles (1974)

- minimalism, autonomy
 - interconnect different networks
- best effort service model
 - simplify routers
- stateless routers
 - keep complexity at the edge (hosts)
- decentralized control
 - scalability

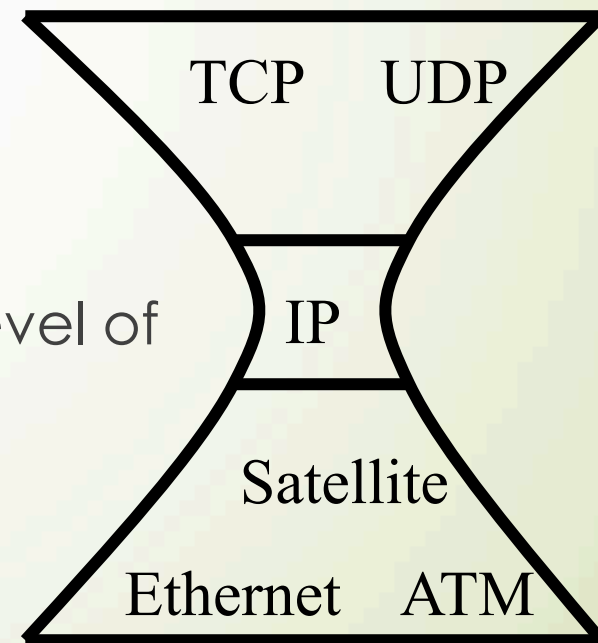
Design Principles Learned from the Internet History

- **Modularity** (e.g., layering)
 - how to break network functionality into modules
- **End-to-End Argument**
 - where to implement functionality
- Separating **policies** from **mechanisms**
 - decouple control from data
 - another form of modularization
- Design for **scale**
 - hierarchy, aggregation, etc.

A Packet-Switched Datagram Network with IP as the **Compatibility** Layer

A prioritized list

1. Connect existing networks
 - initially ARPANET and ARPA packet radio network
2. Survivability
 - ensure communication service even with network and router failures
3. Support multiple types of services
4. Must accommodate a variety of networks
5. Allow distributed management
6. Allow host attachment with a low level of effort
7. Be cost effective
8. Allow resource accountability



Wish List for Today (Future?) Internet

- **Availability** and **reliability**
 - “Always on”, fault-tolerant, fast recovery from failures, ...
- **Quality-of-service (QoS)** for applications
 - fast response time, adequate quality for VoIP, IPTV, etc.
- **Scalability**
 - billions or more of users, devices, ...
- **Mobility**
 - untethered access, mobile users, devices, ...
- **Security** and **Privacy**
 - Protect against malicious attacks, accountability of user actions
- **Manageability**
 - configure, operate and manage networks
 - trouble-shooting network problems
- **Flexibility, Extensibility, Evolvability,**
 - ease of new service creation and deployment
 - evolvable to meet future needs?

Questions

chsu@cs.nthu.edu.tw

